

CHAPTER VII

DESIGN

7.1 Introduction

The main focus for this chapter is the design of the proposed Information and Communication Technologies Resource Management System for the University of Zimbabwe. The previous chapter analysed the existing system and showed that the current environment is fragmented, semi-manual, and weak in asset tracking, maintenance monitoring, reporting, and accountability. The analysis also concluded that the most suitable solution is the development of a centralized system that can register ICT assets, capture faults, monitor maintenance, support allocation, and generate management reports.

The purpose of this chapter is to show how the proposed system will work and how its components fit together. This chapter covers the proposed system design, architectural design, physical design, database design, program design, interface design, pseudo code, security design, and chapter conclusion.

The design of the proposed system is based on the requirements identified in Chapter 6 and the study objectives stated earlier in the dissertation. The system is designed to address the weaknesses of the current environment while remaining practical, manageable, and relevant to the University of Zimbabwe context.

7.2 System Design

The proposed system is a centralized web-based ICT resource management and monitoring system. Its main purpose is to provide one integrated platform through which the University of Zimbabwe can register, track, monitor, allocate, maintain, and report on ICT resources. The system is intended to reduce fragmentation, improve visibility, and support better decision making. This is in line with the study objectives, which include designing and developing a centralized ICT resource management system for the university.

The system will work through role based access. Different users will log in and interact with the system according to their responsibilities. End users will mainly report faults and view updates. ICT administrators will manage asset records, maintenance records, and user accounts. Department representatives will request or track assets assigned to their departments. University management will view dashboards and reports. Suppliers or technicians may receive service requests and provide maintenance feedback where necessary.

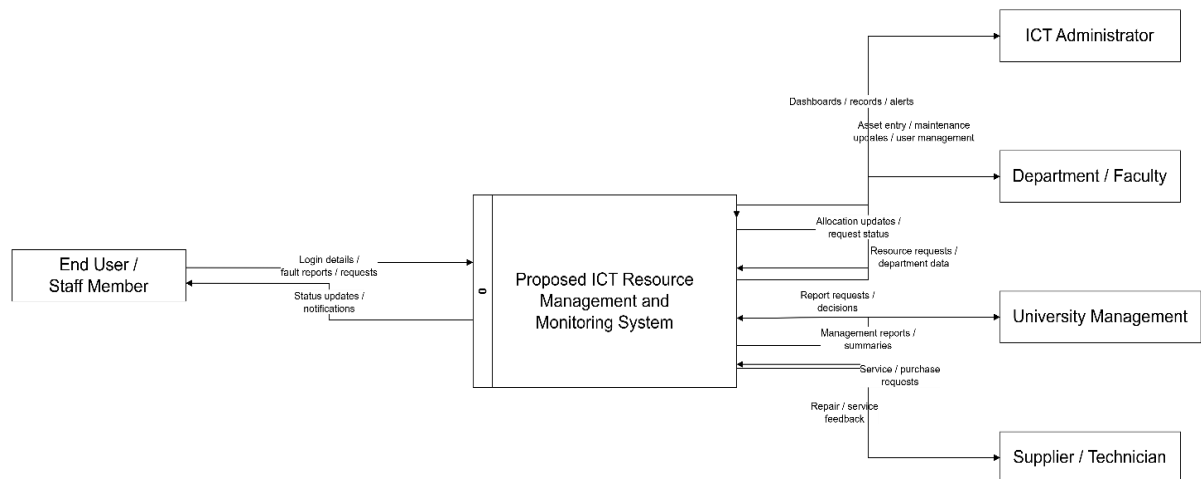
The main functional parts of the proposed system are:

1. user login and authentication
2. asset registration and update
3. fault reporting
4. maintenance monitoring
5. resource allocation and transfer tracking
6. dashboard and report generation
7. user and role management

These functions were derived from the weaknesses of the current system and the requirements already identified in the previous chapters.

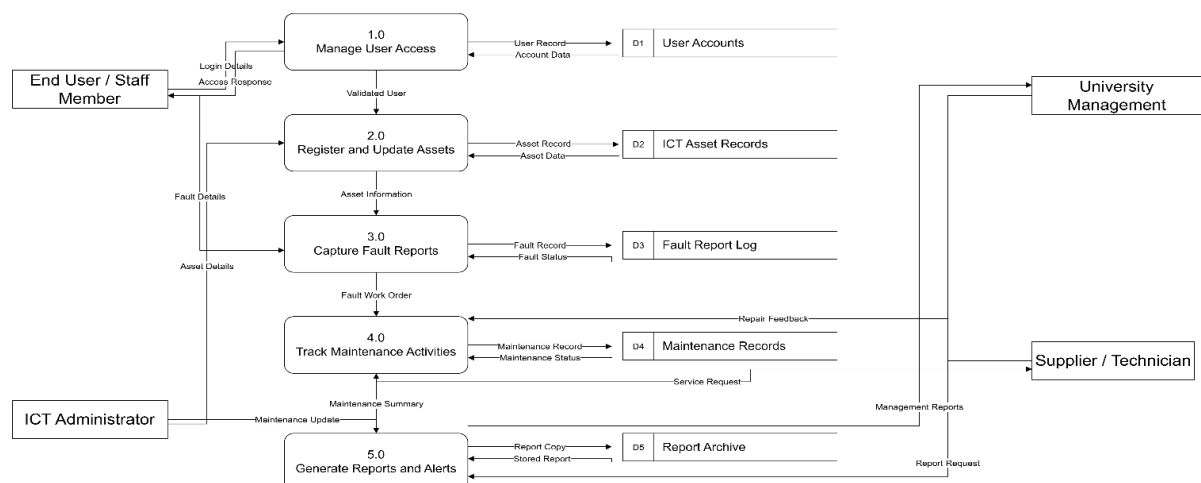
7.2.1 Context Diagram and DFD of the Proposed System

Proposed System Context Diagram



The context diagram shows the proposed system as one central process interacting with the major external entities. This reflects the move from a fragmented environment to one integrated platform for ICT resource management.

Proposed System DFD (Level 0)



This DFD shows the main internal processes of the proposed system and how data is stored and used. The design is consistent with the proposed system already developed in earlier chapters.

7.3 Architectural Design

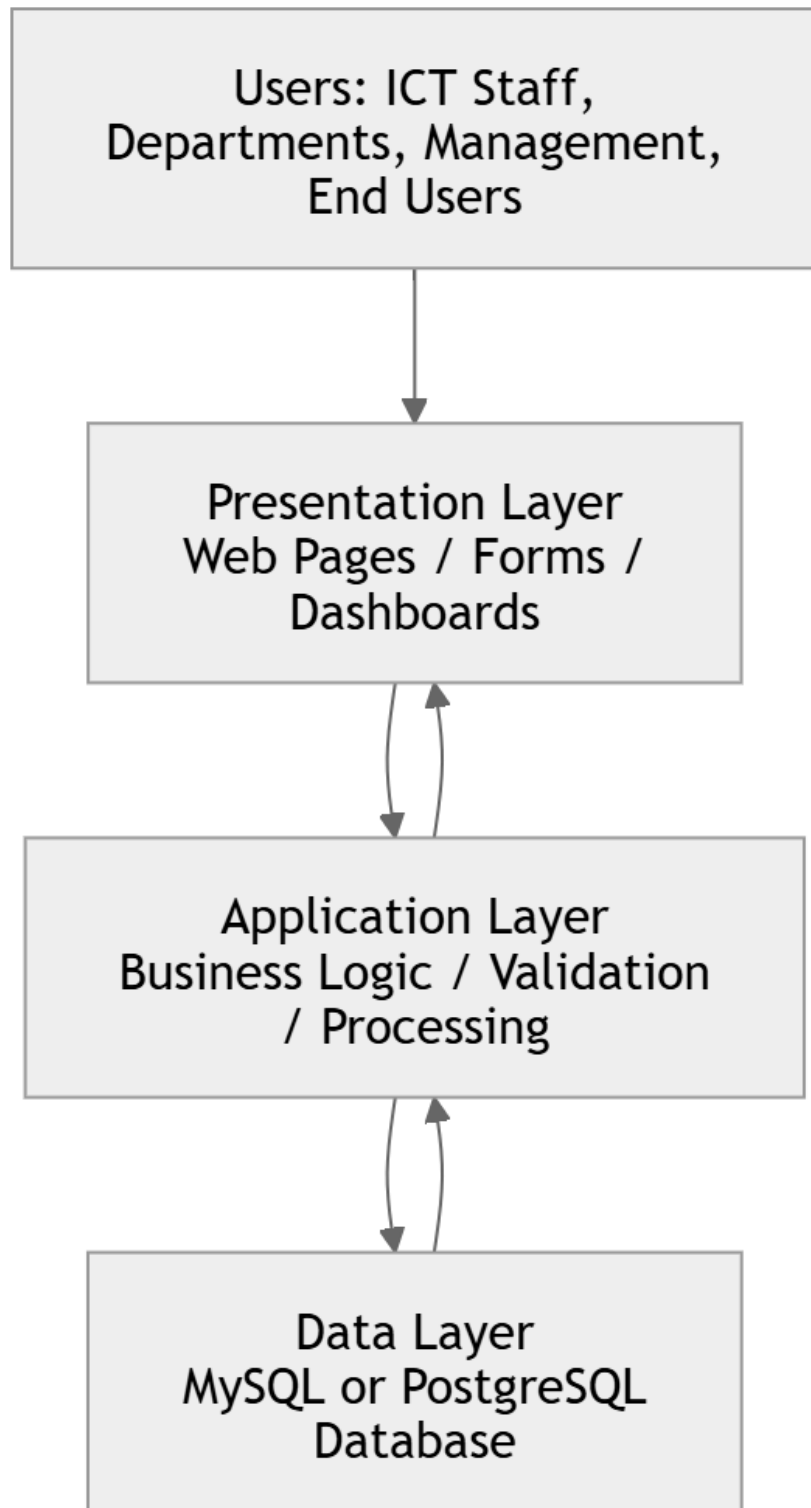
The proposed system will use a three-tier architecture. This architecture separates the system into the presentation layer, application layer, and data layer. A three-tier design is suitable because it improves organization, maintainability, and scalability in web-based systems (Dennis, Wixom and Tegarden, 2015).

The first layer is the **presentation layer**. This is the user interface that users interact with through a web browser. It includes login pages, forms, dashboards, menus, and reports.

The second layer is the **application layer**. This contains the logic of the system. It handles user authentication, asset registration, maintenance processing, allocation processing, report generation, and all business rules of the system.

The third layer is the **data layer**. This is the database layer where information on users, ICT assets, faults, maintenance records, departments, and allocations is stored.

Architectural Design Diagram



This architecture is suitable because it separates user interaction from system processing and from data storage. It also allows easier maintenance and future system expansion.

7.4 Physical Design

Physical design explains how software and hardware will interact in the proposed system. The system will operate as a web-based platform. Users will access it from computers connected to the university network or internet. The system application will run on a server or a hosted machine, and the database will store and manage all records centrally.

The hardware components will include:

- client computers or laptops for users
- a server or deployment computer
- network infrastructure such as routers and switches
- backup power devices where necessary

The software components will include:

- operating system
- web server environment
- application software
- database management system
- browser-based user interface

Physical Design Diagram



The physical design supports centralized access and centralized storage of ICT resource information. This is important because the current system lacks a single integrated point of control.

7.5 Database Design

Database design is important because it provides a structured way of organizing data so that it can be stored, retrieved, and updated efficiently and accurately (Silberschatz et al., 2019; Elmasri and Navathe, 2017). The database design is based on the major data entities identified through the analysis and proposed DFDs. These include users, departments, ICT assets, faults, maintenance records, and allocations.

The major tables in the system are:

- **Users**
- **Departments**
- **ICT_Assets**

- **Fault_Reports**
- **Maintenance_Records**
- **Allocations**

A simple structure of these tables is shown below.

Users Table

- user_id
- full_name
- username
- password
- role
- department_id
- email
- status

Departments Table

- department_id
- department_name
- faculty_name

ICT_Assets Table

- asset_id
- asset_name
- category
- serial_number
- department_id
- location
- date_acquired
- status
- assigned_user

Fault_Reports Table

- fault_id
- asset_id
- reported_by
- report_date
- fault_description
- fault_status

Maintenance_Records Table

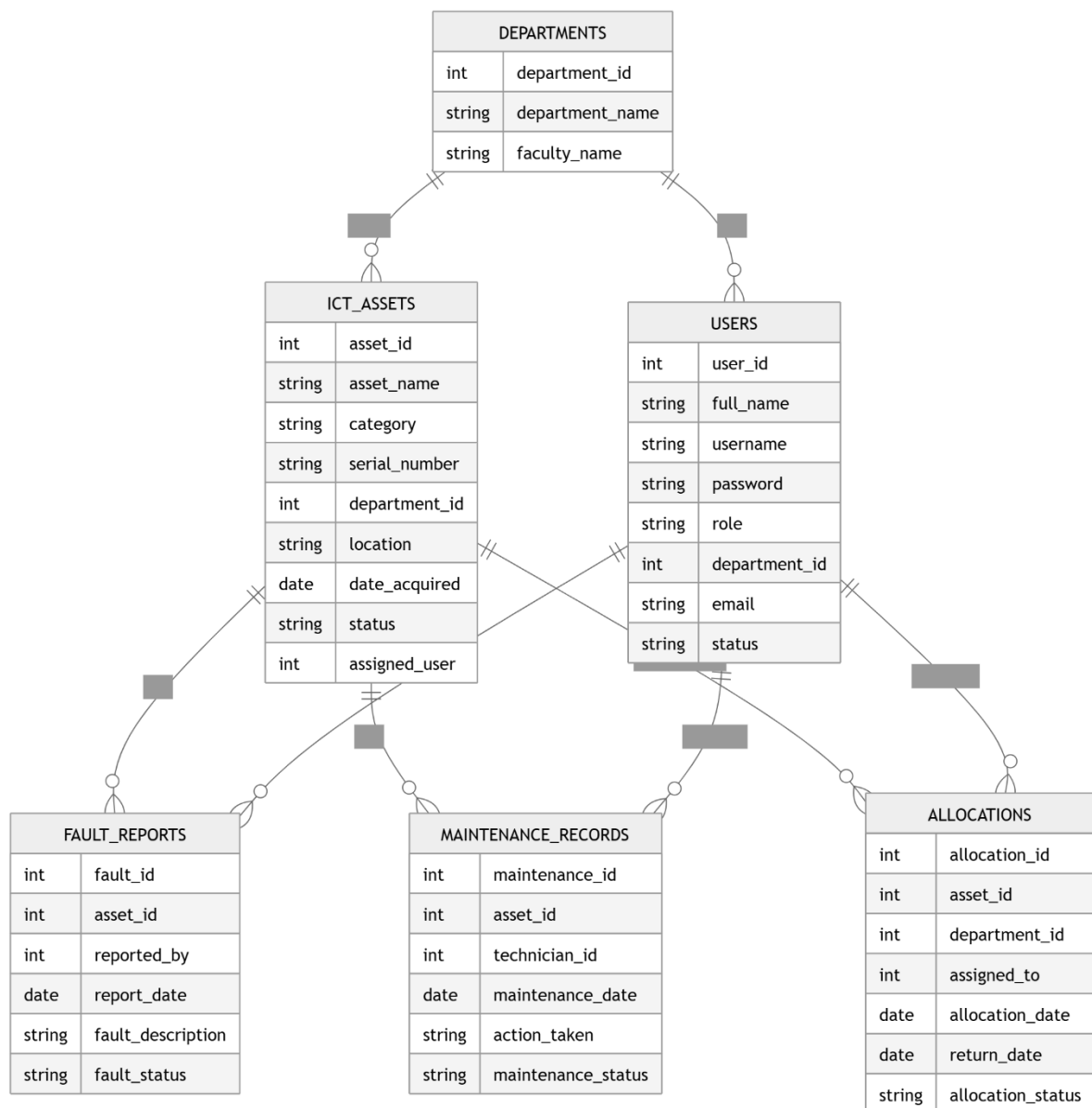
- maintenance_id
- asset_id
- technician_id
- maintenance_date

- action_taken
- maintenance_status

Allocations Table

- allocation_id
- asset_id
- department_id
- assigned_to
- allocation_date
- return_date
- allocation_status

EER / Table Relationship Diagram



This database design supports the main functions of the proposed system and gives a structured way of storing and retrieving information.

7.6 Program Design

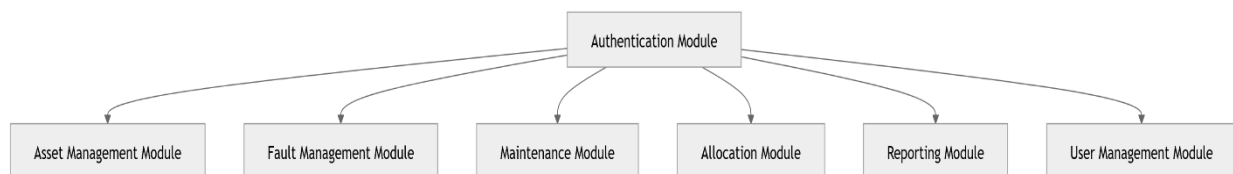
Program design explains how the software components of the system will be arranged and how system objects or modules will interact.

A modular approach is suitable for this system. The proposed modules are:

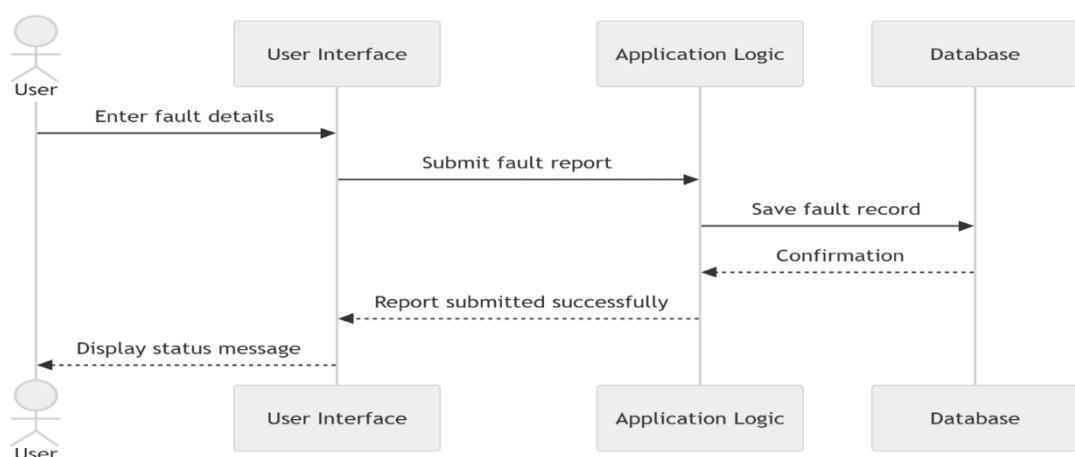
- Authentication module
- Asset management module
- Fault management module
- Maintenance module
- Allocation module
- Reporting module
- User management module

This modular structure improves maintainability because each part of the system handles a specific responsibility.

Package / Module Diagram



Sequence Diagram for Fault Reporting



This sequence shows how one of the core functions, fault reporting, will work in the system.

7.7 Interface Design

A well-designed user interface should be simple, consistent, and easy to understand, since usability strongly affects user acceptance and overall system effectiveness (Rogers et al., 2023; Nielsen, 1994). Interface design focuses on how users will interact with the proposed system. The interface should be clear, simple, and easy to navigate because one of the goals of the study is to improve usability and encourage acceptance of the system. This is also consistent with the user acceptance issues discussed in Chapter Two.

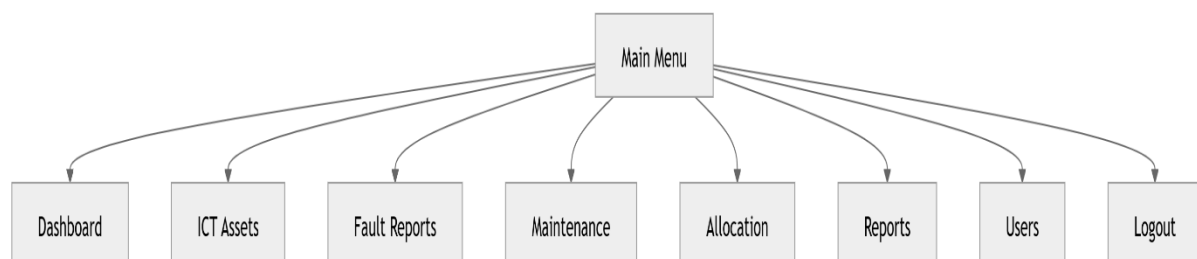
7.7.1 Menu Design

7.7.1.1 Main Menu

The main menu of the system may include:

- Dashboard
- ICT Assets
- Fault Reports
- Maintenance
- Allocation
- Reports
- Users
- Logout

Main Menu Diagram



7.7.1.2 Sub-Menus

Examples of sub-menus include:

ICT Assets

- Add Asset
- View Assets
- Update Asset
- Search Asset

Fault Reports

- Report Fault
- View Faults
- Update Fault Status

Maintenance

- Record Maintenance
- View Maintenance History

Reports

- Asset Report
- Fault Report
- Maintenance Report
- Allocation Report

7.7.2 Input Design

Input design refers to forms through which data enters the system. The main input forms will include:

- login form
- asset registration form
- fault reporting form
- maintenance update form
- allocation form
- user registration form

The forms should include validation so that incomplete or incorrect data is not accepted.

Example Input Form Fields

Asset Registration Form

- Asset name
- Category
- Serial number
- Department
- Location
- Date acquired
- Status
- Assigned user

Fault Reporting Form

- Asset ID
- Reported by
- Date

- Fault description
- Status

7.7.3 Output Design

Output design refers to the information produced by the system. The major outputs of the proposed system will be:

- asset register report
- faulty assets report
- maintenance report
- allocation report
- dashboard summary
- user activity summary

These outputs are important because earlier chapters established that the current system is weak in timely and accurate reporting.

7.8 Pseudo Code

Pseudo code is used to describe the logic of system functions in a simple step-by-step form.

Pseudo Code for Asset Registration

```
START
DISPLAY asset registration form
INPUT asset details
IF any required field is empty
    DISPLAY error message
ELSE
    SAVE asset details into ICT_Assets table
    DISPLAY success message
END IF
STOP
```

Pseudo Code for Fault Reporting

```
START
DISPLAY fault reporting form
INPUT fault details
CHECK if asset exists
IF asset does not exist
    DISPLAY "Invalid asset"
ELSE
    SAVE fault report into Fault_Reports table
    DISPLAY confirmation message
END IF
STOP
```

Pseudo Code for Report Generation

```
START
SELECT report type
RETRIEVE relevant records from database
```

```
FORMAT report output
DISPLAY report on screen
IF user selects print
    SEND report to printer
END IF
STOP
```

7.9 Security Design

Security is important because the system will store institutional data about ICT resources, faults, maintenance, and allocations. The design of the system must therefore provide protection against unauthorized access, misuse, and loss of data.

7.9.1 Physical Security

Physical security refers to the protection of the hardware and facilities used by the system. The server or hosting computer should be kept in a secure room with controlled access. Backup devices should also be stored safely. This reduces the risk of theft, physical damage, or unauthorized physical access.

7.9.2 Network Security

Network security focuses on protecting the system while data is being transmitted over the network. The system should use user authentication, strong passwords, secure connections, and access restrictions. Firewalls and secure network configurations should be used to reduce unauthorized access and attacks over the network.

7.9.3 Operational Security

Operational security refers to procedures that guide the daily safe use of the system. These include:

- role-based access control
- regular password changes
- data backup procedures
- audit trails for important actions
- training users on proper system use
- limiting users to functions relevant to their roles

The security design is important because your earlier chapters already identified confidentiality and accountability as important issues in the study.

7.10 Conclusion

This chapter presented the design of the proposed ICT Resource Management System for the University of Zimbabwe. The chapter explained how the system will work and presented the proposed system context diagram and DFD. It also described the architectural design, physical design, database structure, program modules, interface design, pseudo code, and security design.

The design shows that the proposed system is organized around the core needs identified in the earlier chapters, namely asset registration, fault reporting, maintenance monitoring, allocation control, and reporting. By using a centralized design, the system is expected to improve ICT resource visibility, accountability, usability, and decision-making within the university. The next chapter will focus on implementation and testing of the proposed system.

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